

Ethical work climate, whistleblowing system and their influence on organizational commitment among the Malaysian public sector auditors

Chew Har Loke, Suhaiza Ismail and A.H. Fatima
*Department of Accounting, International Islamic University Malaysia,
Kuala Lumpur, Malaysia*

Abstract

Purpose – The purpose of this paper are twofold. Firstly, it explores the extent to which ethical work climate (EWC) is prevalent among public sector auditors at the National Audit Department (NAD) of Malaysia. Secondly, it examines the relationship between EWC, whistleblowing system and the organizational commitment of public sector auditors in the NAD of Malaysia.

Design/methodology/approach – This study used a postal questionnaire survey, which was distributed to the public sector auditors in Malaysia. A total of 823 responses from public sector auditors in 213 audit teams was received, constituting a 66% and 78% response rate, respectively. To achieve the first research objective, the responses were analysed using descriptive statistical analysis, including mean values and mean value ranking. For the second objective, hierarchical linear modelling (HLM) statistical software was used for the cross-level data analysis.

Findings – The results reveal a moderate level of the overall EWC at the NAD. The overall EWC at the team level also has a positive significant influence on organizational commitment assessed at the individual level of auditors. In terms of the individual components of the EWC, multivariate regression analysis of HLM shows collective moral (CM) sensitivity, CM judgement and whistleblowing system have a significantly positive relationship with organizational commitment, whereas CM motivation has a negatively significant relationship with organizational commitment.

Practical implications – The empirical evidence on the level of EWC and its components implies that there is room for improvement in the ethical climate in the NAD. The finding is important for the NAD to design necessary measures to enhance the EWC, including tightening the auditors' code of ethics and organizing ethics-related seminars, training and workshops for the auditors. The result on the positive influence of CM sensitivity, CM judgment and whistleblowing system on organizational commitment of public sector auditors is also crucial for the NAD to further inculcate moral sensitivity and moral judgment and strengthen the whistleblowing system among the public sector auditors.

Originality/value – This study is among the initial research that uses Arnaud's EWC model in assessing the influence of EWC and whistleblowing system on organizational commitment, particularly in the context of public sector organization of a developing country, i.e. Malaysia.

Keywords Malaysia, Organizational commitment, Ethical work climate (EWC), Public sector auditors, Cross-level

Paper type Research paper



1. Introduction

Recent ethical scandals involving auditors have raised concerns about the credibility of the accounting profession. Ethical scandals occur not only in the private sector but also in the public sector. When it involves public sector auditors, the implication is often more

detrimental, with a huge negative impact on society and the country. This is because public sector auditors have been entrusted to ensure public resources are being used efficiently to better serve the public at large. Ethical scandals in the public sector eventually lead to public dissatisfaction and distrust of the government, which may cause disharmony among society, which subsequently may adversely affect the economic development of the country (Cameron, 2004; Abdullah *et al.*, 2014).

An example would be the case of the Youth and Sports Ministry of Malaysia involving embezzlement of a huge sum of public funds (*The Malay Mail*, 2016), caused by the abuse of the standard operating procedures for procurement (*The Straits Times*, 2016). The Securities Commission's subsequent investigation of the group involved found that revenue figures had been falsified. More recently, in relation to the world-widely known 1Malaysia Development Berhad case, the audit firm partnership was sued due to allege audit breaches and negligence in failing to prevent the large-scale misappropriation (Consultancy.asia, 2021). These evidences raised the issue of why the auditor (in the above cases, Klynveld Peat Marwick Goerdele, which was appointed on behalf of the National Audit Department [NAD]), had failed to detect or to highlight the embezzlement.

The NAD is a public sector organization in Malaysia entrusted with performing an audit on all other public sector organizations of the country to ensure that they properly discharge their accountability to the public. Hence, the NAD holds a vital responsibility in auditing and reporting the performance of public sector organizations from different aspects. The former Auditor General (AG) of Malaysia, Tan Sri Dato' Setia Haji Ambrin Bin Buang (2010), confirmed that "auditor and the audit work are significant elements in reducing fraud and misconduct". Thus, the ethics of the public sector auditors of the NAD is important as they are accountable to the public.

Prior studies have reported that one factor that influences the organizational commitment of individuals is ethical work climate (EWC). The EWC is one of the important concepts of ethics and refers to a shared perception of organizational norms, values and behaviour (Schneider, 1975; Simha and Cullen, 2012). The essence of having a strong EWC is to disseminate organizational ethical values to the members for their daily work activities. It is expected to shape organizational members' organizational commitment (Pritchard and Karasick, 1973; Lemmergaard and Lauridsen, 2007; Barattucci *et al.*, 2021). Having an appropriate level of EWC is essential as it may influence the organizational commitment of an individual in an organization (Victor and Cullen, 1988; Cohen, 1995; Kish-gephart *et al.*, 2010; Pagliaro *et al.*, 2018; Teresi *et al.*, 2019).

Realising the importance of ethics and EWC in a public sector organization and the limited research on EWC in developing countries, the present study explores the EWC in the NAD in Malaysia as perceived by the public sector auditors. Using Arnaud's EWC model (2010), the present study covers both the overall EWC as well as its four components. The present study also examines the association between EWC, whistle-blowing system and the organizational commitment of public sector auditors in Malaysia. The contributions of this study are that it assesses the level of ethics of the public sector auditors in the NAD in Malaysia, using the newly developed EWC model by Arnaud (2010) and uses a multi-level approach to measure the shared perception of the members of the EWC of the organization, which to date, has been used by very few studies.

The remaining sections of this study are organized as follows. Section 2 reviews relevant literature in the context of the present study and presents the theoretical framework and hypotheses of this study. Then, Section 3 discusses the research

methodology and Section 4 presents the research findings. Finally, Section 5 offers the implications of the study, outlines its limitations and suggestions for future studies and provides a conclusion.

2. Theoretical framework, literature review and hypotheses development

2.1 Theoretical framework

The theoretical framework of the present study is developed based on three models. The first model is [Arnaud's \(2010\)](#) psychological process model (PPM), which was developed from [Rest's \(1994\)](#) four psychological processes ethical decision-making model (EDMP) (second model) that consists of moral sensitivity, moral judgment, moral motivation and moral character and the third model is [Hunt and Vitell's \(1986, 1991\)](#) general theory of marketing ethics. [Rest \(1994\)](#) believed that these four individual moral components to assess decision-making would lead to ethical conduct/behaviour. In fact, "moral failure can occur because of deficiency in any component" ([Rest, 1994](#), p. 24). [Rest \(1994\)](#) further explained that there are interactions among the four components, and it is not necessary to follow the sequence of components; however, "the four components comprise a logical analysis of what it takes to behave morally" (p. 24).

[Arnaud \(2010\)](#) transformed [Rest's \(1994\)](#) model to become the EWC model as a multi-level model. [Arnaud's \(2010\)](#) EWC model is collectively conceptualized into four psychological processes: collective moral (CM) sensitivity, CM judgment, CM motivation and whistleblowing system. CM sensitivity entails two sub-components – norms of moral awareness and norms of empathetic concern. Likewise, CM judgment consists of two sub-components – focus on self and focus on others. Taken together, there are six components in [Arnaud's \(2010\)](#) model.

CM sensitivity refers to considering possible ethical actions (awareness) and assessing the impact of consequences on others at the group-level perspective (moral concern). The achievement of a higher level of moral sensitivity is expected to generate more ethical conduct. In addition, a lower level of moral sensitivity may lead the employees to commit unethical acts. At the CM judgement level, this is the right time to apply prevailing norms to assess morally right actions at a group level. A morally right choice will be selected among the possible ethical acts after considering the impact on oneself and others. If an organization is capable of making a morally right choice, then it will lead the organizational members to ethical decision-making. At the CM motivation stage, it shows whether a moral value of an organization is given priority over other values while making ethical decisions. Other values of concern are authority power, personal status, biasness or super achievement. This stage could reveal whether moral values or other values are the preferable habits in decision-making.

The final component of the PPM is whistleblowing system. Organizational members consistently implement a planned course of action based on the norms of self-control or norms for assuming responsibility. Self-control reflects controlling ability in undertaking a planned action. In this case, moral value is always the first consideration. The organizational members also consider their responsibility:

- to others' welfare;
- to maintaining moral commitment; and
- to underlying rules while taking action.

Overall, if an organization attains a positive evaluation of this process, it shows that the organizational members are capable of solving problems ethically. This implies a positive or high EWC exists at a social level. Arnaud's (2010) EWC is measured using the instrument called the ethical climate index (ECI). Hunt and Vitell's (1986, 1991) ethical decision-making model covers the outcomes of ethical decision-making processes; thus, it is applied to the research framework of the study. Figure 1 presents the theoretical framework of this study.

2.2 Literature review and hypotheses of the study

Because ethical scandals are committed by people who fail to behave ethically, research on organizational commitment has gained prominence. In such research, moral development has been a topic of interest since the mid-1950s (Kohlberg, 1968), and this interest continues to date. The early EDMP models were gradually developed and built on, and they were well received among researchers, who recognized that these models could promote better human behaviour (Ferrell and Gresham, 1985; Hunt and Vitell, 1986; Trevino, 1986; Dubinsky and Loken, 1989; Jones, 1991; Rest and Narvaez, 1994; Thomas, 2002).

Generally, prior studies on EWC can be categorized into three:

- (1) studies on the factors influencing EWC (Luthar *et al.*, 1997; Schminke *et al.*, 2005; Weeks *et al.*, 2006; Demirtas and Akdogan, 2015; Newman *et al.*, 2017; Loke, 2019; Otaye-Ebede *et al.*, 2020);
- (2) studies on the outcomes of EWC (Ismail, 2015; Shafer, 2015; Tsai and Huang (2008); Hsieh *et al.*, 2016; Loke *et al.*, 2019; Pagliaro *et al.*, 2018; Teresi *et al.*, 2019; Friend *et al.*, 2020; Otaye-Ebede *et al.*, 2020; Potipiroon and Wongpreedee, 2020; Barattucci *et al.*, 2021); and
- (3) studies on the level or types of EWC (Wittmer and Coursey, 1996; Brower and Shrader, 2000; Rasmussen *et al.*, 2003; Armstrong and Francis, 2008; Loke *et al.*, 2019; Wan Abdul Razak and Mohd Mustamil, 2020).

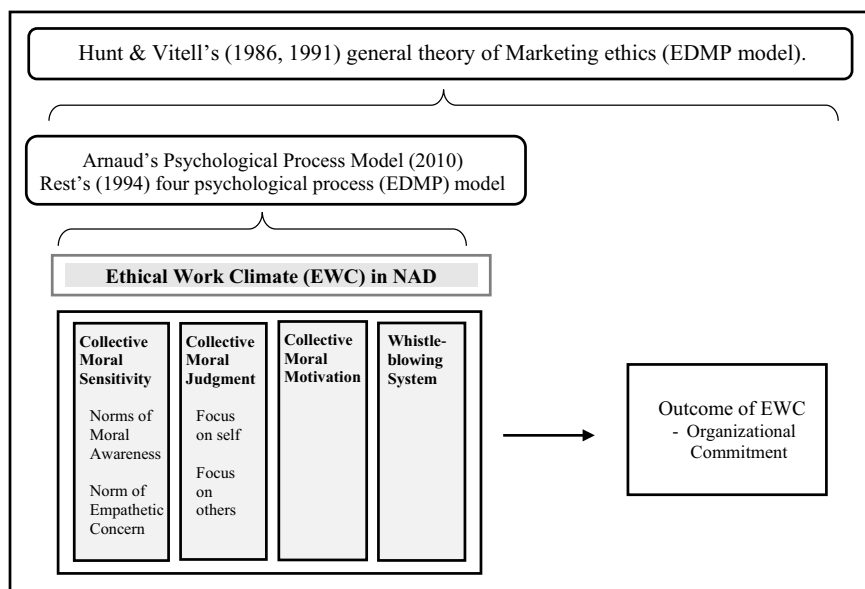


Figure 1.
Theoretical
framework of ethical
work climate in
Malaysian NAD

Prior studies on EWC have adopted either the qualitative or quantitative approaches. [Rasmussen et al. \(2003\)](#) and [Armstrong and Francis \(2008\)](#) used the qualitative approach to investigate the EWC in the Canadian public setting and the Australian private sector, respectively. [Rasmussen et al. \(2003\)](#) conducted face-to-face unstructured interviews of mid-level managers, 12 each from government and non-profit organizations (health and social service sectors) in a province in Canada. The study found differences in EWC in the two organizations. The Canadian public servants were ethically concerned with the public and were guided by EWC with professional legal norms and codes, whereas non-profit organization members were focused more on self-benefit and tended to follow their own principles.

[Armstrong and Francis \(2008\)](#) provided a conceptual discussion of EWC. They urged the private sector's organizational boards to encourage an appropriate EWC in their organization because the board directors are the key personnel who should lead subordinates to act ethically to minimize unethical conduct. They also stated that corporate ethical scandals are partly due to a lack of care by employees. For example, the board may have failed to control organizational ethical practices among the employees. The authors suggested if an ethical work environment can be created to guide employees' ethical behaviour, ethical misconduct may be reduced. Prior studies have also examined the differences in EWC between public and private sector organizations ([Wittmer and Coursey, 1996](#); [Brower and Hrader, 2000](#)). The results reveal significant differences in EWC between the two entities.

Prior studies on the association between EWC and human behaviour can be grouped into two main categories. The first category covers prior studies that have adopted [Victor and Cullen's \(1988\)](#) EWC model, while the second category includes studies that have used [Arnaud's \(2010\)](#) EWC model. In relation to the first category, some studies have analysed the EWC variable at a single level of analysis, whereas other studies have used a multi-level analysis. [Vardi \(2001\)](#) examined both organizational climate (warmth, support and reward) and EWC as the antecedents of organizational misbehaviour. The study also compared the two different climates to ascertain which climate has a more significant influence on organizational misbehaviour. The empirical findings indicate that both climates are statistically significant and have a negative relationship with organizational misbehaviour. [Peterson \(2002\)](#) investigated the association between EWC (egoism, benevolence and principle ethical perspectives) and unethical behaviour using a questionnaire survey. The findings show that the egoism ethical perspective has a significantly positive relationship with unethical behaviour, whereas the benevolent and principle ethical perspectives are significantly and negatively related to unethical behaviour.

A meta-analysis by [Kish-gephart et al. \(2010\)](#) found that the ethical criterion of egoism positively influences unethical behaviour, but the benevolent ethical type and principle ethical type are negatively related to unethical behaviour. These results support [Victor and Cullen's \(1988\)](#) EWC model. [Wimbush et al. \(1997\)](#) examined the relationship between EWC and human behaviour using [Victor and Cullen's \(1988\)](#) model and analysed their data at two levels: individual and district level. The results at the individual level (employees, at level 1) indicate that there is a significantly negative relationship between EWC dimensions – caring, service, laws and codes and independence and unethical behaviour (complicit, disobedient, lying and stealing). As for the result at the district level (district, at Level 2), data was analysed using the within and between analysis technique. The results at the district level (Level 2) show only one significant positive relationship, i.e. between instrument ethical type and unethical behaviour.

In another multi-level study, [Shin \(2012\)](#) studied climate strength as the moderator in the relationship between EWC (focused only on principles (rules) and laws and codes dimension) and organizational citizenship behaviour. The study aggregated the variables of climate strength, EWC and organizational citizenship at the firm level, and the data was analysed using hierarchical regression analysis. The finding demonstrates that climate strength significantly moderates the relationship between EWC and organizational citizenship behaviour. [Arnaud and Schminke \(2012\)](#) explored the extent to which EWC interacts with and influences organizational commitment in the USA. In addition, the study examined the moderating effects of CM emotion and collective ethical efficacy on the relationship between EWC and organizational commitment.

Based on the above review, although studies on both private and public sectors are available, studies specifically on the audit profession in the public sector are lacking. Furthermore, prior studies have measured EWC at a single-level perspective and have been conducted in developed countries. Also, previous studies have tested the types of organizational EWC using [Victor and Cullen's \(1988\)](#) EWC model. Very few studies have used Arnaud's EWC model. [Loke et al. \(2019\)](#) used [Arnaud's \(2010\)](#) ECI to measure the extent of EWC in the Malaysian NAD as perceived by the public sector auditors. Although [Loke et al. \(2019\)](#) confirmed that [Arnaud's \(2010\)](#) ECI can be applied in the context of public sector auditors, they did not empirically evaluate the EWC. Thus, this study fills the gap by conducting empirical research to evaluate the extent of EWC in the NAD as perceived by the public sector auditors.

Furthermore, most prior studies that have examined the relationship between EWC and organizational commitment have used [Victor and Cullen's \(1988\)](#) model to measure EWC, except for [Arnaud \(2010\)](#), who investigated the EWC-organizational commitment relationship using the [Arnaud \(2010\)](#) model, and discovered that only three (CM sensitivity, CM motivation and whistleblowing system) out of four ethical climate components significantly and positively influence organizational commitment. [Arnaud \(2010\)](#), therefore, called for more studies to apply [Arnaud's \(2010\)](#) EWC model to examine its influence on organizational commitment. Only a few studies have analysed EWC using multi-level analysis, specifically in the context of public sector organizations. Therefore, by using [Arnaud's \(2010\)](#) EWC model and adopting a multi-level approach, this study examines the influence of EWC and whistleblowing system on the organizational commitment of public sector accountants. Based on this discussion, the following are hypothesized:

- H1.* There is a positive association between overall EWC at the team level and organizational commitment at an individual level in the NAD.
- H2.* There is a positive association between CM sensitivity and organizational commitment at an individual level in the NAD.
- H3.* There is a positive association between CM judgment and organizational commitment at an individual level in the NAD.
- H4.* There is a positive association between CM motivation and organizational commitment at an individual level in the NAD.
- H5.* There is a positive association between whistleblowing system and organizational commitment at an individual level in the NAD.

3. Research methodology

3.1 Respondents, data collection procedures and response rate

This study used a cross-sectional approach to collect data from the population of public sector auditors working in the NAD in Malaysia. Public sector auditors interact with their

team members and undertake assigned audit duties together. An approval letter was obtained from the former AG of Malaysia, Tan Sri Dato' Setia Haji Ambrin Bin Buang, to engage public sector auditors as the respondents of this study. All related information for the distribution of questionnaires, such as the total number of audit teams, team members of each audit team, postal address, contact email and phone number of the representatives of this survey in all offices, were provided by the NAD. Upon receiving this information, phone calls were made to the respective offices to communicate with the representatives before distributing the questionnaire by postal mail. The researcher obtained assistance from the representatives of each audit department office to distribute, collect and return all completed questionnaires, and for this, the representatives were rewarded with a small token.

An email was sent, or a phone call was made to the representative in each of the national audit offices a week after mailing out the questionnaires to obtain acknowledgement of receipt of the questionnaire and his/her understanding of how the questionnaire must be distributed. Each representative in each national audit office received several packets of survey envelopes, depending on the number of audit teams in the national audit office. The representative distributed to audit team leaders this survey envelope containing a numbered set of questionnaires. Then, each audit team leader distributed the set of questionnaires to each team member. A set of questionnaires consists of a copy of the questionnaire for which they do not have to indicate their name, an official letter of support from the then AG of Malaysia, a cover letter explaining that the survey is for educational purposes only and a reply envelope, including concise guidance to complete the questionnaire and return it to the official representative.

A gentle reminder was sent via email or phone call to audit offices that were late in returning the questionnaires by mail. An additional one to two weeks were given to the audit leader to collect the completed questionnaires from his/her team members and put them in a sealed envelope to return to the representative in his/her national audit office. Once the representative from each audit department office had received all the sealed survey packets from the team leaders, the survey packets were inserted into the self-addressed and postage-paid return envelopes and posted to the researcher. Overall, it took about two months to end the data collection from the initial mailing.

In total, 1,252 sets of questionnaires for 213 teams were distributed to all audit offices. A total of 823 public sector auditors from 213 teams participated in this study, consisting of a response rate of 66% (individuals) and 78% (teams). Out of the 823 returned questionnaires, 31 were eliminated because of missing responses to the research variables. A further 27 individual responses were eliminated from the data analysis due to the guideline of having a minimum of three respondents per team (Schneider *et al.*, 1998; Tracey and Tews, 2005; Mayer *et al.*, 2009). Finally, this resulted in 765 public sector auditors and 149 teams as the valid participation at a net response rate of 61% and 70%, respectively. Table 1 presents the response rate of the study. Generally, the approval granted by the then AG contributed to the high response rate of this study.

3.2 Survey instrument and variables measurement

The survey instrument consists of four sections. Section A is on the background of the respondents and comprises both open-ended and closed-ended items. The background information includes gender, age, qualification, organizational tenure, type of audit involved, team tenure and number of team members. Section B relates to EWC. EWC is measured using the questionnaire instrument developed by Arnaud (2010) as it is designed to capture "the true breath of the ethical climate construct" (Arnaud, 2010, p. 347). The original instrument consists of 36 items. However, two items were excluded from the present

								Ethical work climate
No.	NAD offices	Individual			Team			
		Distribution	Responses	%	Distribution	Responses	%	
1	Johor	64	50	78	12	12	100	727
2	Kedah	47	35	75	9	8	89	
3	Kelantan	59	41	70	9	8	89	
4	Melaka	37	36	97	9	9	100	
5	Negeri Sembilan	42	29	70	8	7	88	
6	Pahang	45	44	98	9	9	100	
7	Perlis	26	23	85	4	4	100	
8	Penang	51	33	65	11	8	73	
9	Perak	59	53	90	10	10	100	
10	Sabah	124	101	82	22	21	96	
11	Sarawak	107	91	85	18	17	94	
12	Selangor	52	31	60	9	7	78	
13	Terengganu	53	47	89	10	10	100	
14	Head office, Putrajaya	486	209	43	73	35	48	
Grand total		1,252	823	66	213	165	78	Table 1. Individual and team response rates
After data cleaning			765	61			70	

study because they had a similar meaning, especially in the context of the current respondents. These two items belong to the sub-component of CM sensitivity, i.e. norms of empathetic concern. The remaining 34 items could still comprise the four main components of EWC: CM sensitivity (with sub-scale: norms of moral awareness and norms of empathetic concern) (10 items), CM judgement (10 items), CM motivation (8 items) and whistleblowing system (6 items). Because EWC is measured at the team level, the items were worded to reflect a team perspective – for example, “Auditors in my audit team feel it is better to assume responsibility for a mistake”, “In my audit team, personal success is more important than helping others” and “When necessary, auditors in my audit team take charge and do what is morally right”. Table 2 shows Cronbach’s alpha values of EWC and its components are above 0.80, confirming that the data of the current study is reliable and consistent.

The organizational commitment variable is presented in Section C of the questionnaire. There are nine items on organizational commitment obtained from the studies of Kaptein and Avelino (2005) and Park and Blenkinsopp (2013). All survey items were anchored on a seven-point Likert scale, ranging from “Strongly Disagree” (1) to “Strongly Agree” (7). The lower scale means weaker EWC and organizational commitment, whereas the higher scale indicates stronger/more positive commitment. Compared with a five-point Likert scale, a seven-point Likert scale provides more choices of scale, which helps to reduce sensitivity in answering the questionnaire, specifically in answering the negative items. In addition, from the data analysis perspective, more choices of scale

Variables	Cronbach’s alpha	No of items	Table 2. Reliability statistics for ethical work climate and its components
Ethical work climate (EWC)	0.93	32	
EWC-component			
Collective moral sensitivity	0.89	8	
Collective moral judgment	0.81	10	
Collective moral motivation	0.91	8	
Whistleblowing system	0.84	6	

results produce a more refined measurement, enabling a more detailed data analysis to be performed. The questionnaire instrument was in both English and Malay languages. The original questionnaire was in the English language; it was translated to the Malay language by the Malaysian Institute of Translation and Books (*Institut Terjemahan and Buku Malaysia*).

3.3 Pilot tests

Two pilot tests were conducted before the actual distribution of the questionnaire. Eleven lecturers from the International Islamic University Malaysia Department of Accounting and the former Director of the Special Audit and Research Sector in the NAD, Dr Masiah, participated in the first pilot study. The purpose of this pilot test was to ensure the relevance of the questions and to verify the content validity of measuring each variable in the questionnaire. Overall, the respondents pointed out some vague items, time spent to complete a set of the survey and minor grammatical mistakes. Comments given by the respondents are fully appreciated, and necessary amendments were made to improve the instruments.

Subsequent to improving the questionnaire based on the first pilot test, the second pilot test was performed on private-sector auditors. Specifically, questionnaires were distributed to private-sector auditors in four small, medium and large accounting firms in West Malaysia. The reason for selecting private sector auditors as the respondents for the second pilot test is because they are also practising auditors and, therefore, likely to face similar ethical work-related issues as the actual respondents in the study.

The reasons for conducting the second pilot test are to further enhance content validity, to give further assurance on the readability, understandability and appropriateness of the questionnaire items, and more importantly, to ensure the reliability of the instrument. For this study, the reliability of the questionnaire instrument is more convincingly verified from the findings of the second pilot test, as the respondents are practising auditors.

3.4 Data analysis

The present study used a statistical package for social sciences, version 20, to perform descriptive analysis on the demographics and variables data. Descriptive statistics, including minimum and maximum values, mean, frequency and percentage values, were calculated. To examine the level of EWC, mean scores and mean score ranking were used. To examine the hypotheses of the study, hierarchical linear modelling (HLM) software Version 7 was used. Multivariate regression of HLM was specifically used to examine the cross-level direct effects of EWC and whistleblowing system on organizational commitment in the NAD – the EWC was measured at the team level, and outcomes of EWC were measured at an individual level.

[Table 3](#) presents the HLM equations for testing the hypotheses:

3.5 Preliminary tests

The preliminary tests of normality, reliability and multicollinearity test were also carried out to check for violation of assumptions. The normality tests, as shown in [Table 4](#), which include mean, trimmed mean, skewness and kurtosis, indicate that based on 5% trimmed mean, there is no difference between original mean and trimmed mean of respective variables; this implies that data for each variable is normal. With regards to skewness and kurtosis, the distribution of scores was normal for respective variables. In terms of reliability test, all the variables and sub-variables have Cronbach's alpha values of above 0.70 (refer to [Table 2](#)), which indicates the data of the current study is reliable. Likewise, there is no multicollinearity issue for components of EWC, as the results are within the required range (refer to [Table 5](#)).

Table 3.
Hierarchical linear
models for testing
the hypotheses

	Algebraic symbols equation	Variable labels equation	Hierarchical linear models	Where:
Multivariate multilevel regression	Level-1 Model $DV_{ij} = \beta_{0j} + r_{ij}$	Level-1 Model $Organizational\ commitment_{ij} = \beta_{0j} + r_{ij}$		β_{0j} = random intercept
	Level-2 Model $\beta_{0j} = \gamma_{00} + \gamma_{01} * (IV_j) + \gamma_{02} * (IV_j) + \gamma_{03} * (IV_j) + \gamma_{04} * (IV_j) + u_{0j}$	Level-2 Model $\beta_{0j} = \gamma_{00} + \gamma_{01} * (Sensitivity)_j + \gamma_{02} * (Judgment)_j + \gamma_{03} * (Motivation)_j + \gamma_{04} * (Whistleblowing\ system)_j + u_{0j}$		γ_{00} = grand mean
	Mixed model $DV_{ij} = \gamma_{00} + \gamma_{01} * IV_{1j} + \gamma_{02} * IV_{2j} + \gamma_{03} * IV_{3j} + \gamma_{04} * IV_{4j} + u_{0j} + r_{ij}$	Mixed model $Organizational\ commitment_{ij} = \gamma_{00} + \gamma_{01} * Sensitivity_j + \gamma_{02} * Judgment_j + \gamma_{03} * Motivation_j + \gamma_{04} * Whistleblowing\ system_j + u_{0j} + r_{ij}$		u_{0j} = random error term

Table 4.
Statistical tests to
assess normality of
variables

	Statistic	Standard error	Notes
<i>a. Ethical work climate (EWC)</i>			
Mean	5.21	0.026	5.21–5.21 = 0 (data is normal)
5% Trimmed mean	5.21	–	
Standard deviation	0.731	–	
Skewness	–0.334	0.088	Between –3.0 and +3.0
Kurtosis	0.728	0.176	Between –10.0 and +10.0
<i>b. Organizational commitment</i>			
Mean	5.46	0.031	5.48–5.46 = 0.02 (data is normal)
5% Trimmed mean	5.48	–	
Standard deviation	0.864	–	
Skewness	–0.367	0.088	Between –3.0 and +3.0
Kurtosis	–0.358	0.177	Between –10.0 and +10.0

Table 5.
Multicollinearity test:
tolerance value and
variance inflation
factor on EWC

Model	Coefficients ^a	Collinearity statistics	
	Tolerance		VIF
Collective moral sensitivity	0.484		2.065
Collective moral judgement	0.395		2.534
Collective moral motivation	0.564		1.774
Whistleblowing system	0.452		2.212
Note: ^a Dependent variable: ethical work climate			

4. Findings and discussion

4.1 Demographic information of respondents and descriptive statistics

Table 6 presents the demographic information of the respondents.

The majority of respondents (70%) are female, and this result is consistent with the statistics provided by the NAD. In total, about 85% of the participants are over 30 years old, which indicates better ability in terms of handling audit work. Almost 45% of the respondents have more than nine years of audit experience in the NAD, whereas another 34% have between six and nine years of experience. This indicates more experience in managing ethical matters. In terms of qualifications, approximately 51% have a diploma, 44% have a bachelor's degree and approximately 8% have a professional accounting qualification. Generally, public sector auditors stay in the same audit team for about two years unless they request a transfer. The statistics of the study show that about 56% of respondents have remained in the same audit team for more than two years. Overall, the demographic information shows that the respondents have considerable competence and experience as team members to participate in the present study and answer the items in the questionnaire.

Table 7 describes the descriptive statistics of the main variables covered in this study. The mean scores for all the variables are above 5 (i.e. above mid-point of 4), which indicates that, on average, the EWC at the NAD is acceptable and members show organizational commitment at work.

					Ethical work climate
Item	Responses	Frequency (N = 765)	(%)	Total	
Gender	Male	219	28.60	100	
	Female	534	69.80		
	Missing	12	1.60		
Age	<25 years	7	0.90	100	
	25 to < 30 years	105	13.70		
	30 to < 35 years	296	38.70		
	35 to < 45 years	220	28.80		
	≥45 years	130	17.00		
	Missing values	7	0.90		
	≤3 years	41	5.40	100	
Length of work experience in NAD	>3–6 years	120	15.70		
	>6–9 years	262	34.20		
	>9–12 years	70	9.20		
	>12–15 years	103	13.50		
	>15 years	168	22.00		
	Missing values	1	0.10		
Highest academic qualification	Diploma	381	49.80	100	
	Advance diploma	9	1.20		
	Degree	309	40.40		
	Master	36	4.70		
	Others	30	3.90		
	Missing values	0	0.00		
With professional qualification	Yes	59	7.70	100	
	No	691	90.30		
	Missing value	15	2.00		
Types of Audit Team	Financial audit team	520	68.00	100	
	Performance audit team	203	26.50		
	Governance audit team	37	4.80		
	Missing values	5	0.70		
	≤1/2	185	24.18	100	
Duration in the current audit team	>1/2 to 1	58	7.58		
	>1 to 1 1/2	38	5.00		
	>1 1/2 to 2	56	7.30		
	>2	426	55.70		
	Missing values	2	0.03		

731

Table 6.
Demographic of respondents

4.2 Preliminary analysis results prior to running cross-level interaction effect

Public sector auditors work in audit teams; thus, the extent of EWC is measured at the team level. The individual-level measures of EWC of public sector auditors were aggregated at the team level (Level 2) to measure EWC in a collective manner. To justify the suitability of aggregating EWC, within-group interrater agreement (mean $r_{(WG)}(\rho)$), significant between-

Variables	Mean	
Overall EWC	5.19	Table 7. Mean statistics on main variables
Collective moral sensitivity	5.43	
Collective moral judgment	5.07	
Collective moral motivation	5.49	
Whistleblowing system	5.26	
Organizational commitment	5.46	

team (F) and intra-class coefficient (ICC) methods, were assessed (James *et al.*, 1984) for overall EWC and its components. Mathieu *et al.* (2007) believed that mean $r_{(WG) (j)} \geq 0.70$ is suitable to establish variables to a higher level. In this case, EWC was aggregated as the audit team is nested at the upper level (Level 2). The $r_{(WG) (j)}$ index for 149 teams is 0.98 (SD = 0.03) for EWC, which is greater than the suggested guideline. This represents 98% homogeneity level of EWC perception of teams. Thus, EWC data was suitable to be aggregated to a higher level, i.e. at the audit team level. Equally important, the four components of EWC were subjected for reliability ($r_{(WG) (j)}$) test and the results are 0.90 (SD 0.17) for the first component of EWC: CM sensitivity, 0.90 (SD 0.25), for the second component: CM judgment, 0.96 (SD 0.25), 0.77 (SD 0.96) for CM motivation and 0.90 (SD 0.14) for whistleblowing system; the $r_{(WG) (j)}$ findings indicate 90%, 96%, 77% and 90% of homogeneity level of perception among public sector auditors on respective EWC components.

Team measures should also vary significantly between teams. Using one-way random effect ANOVA to measure Level 2 variables, the current study finds that the F -value of EWC is 1.45 ($p < 0.001$), whereas the F values for components of EWC show statistically significant variance as in Table 6. Next, ICC is another measure of between-teams variance. It is 0.078 for EWC, meaning that 7.80% of the variance in EWC is due to team random effects. Based on Table 6, other constructs' variance is below 7% (0.070), which is considered as good. Taken together, the mean $r_{(WG) (j)}$ values, F -values and ICC values justified aggregating EWC and measuring it at the team level in the present study. Table 8 presents the descriptive statistics, correlations between variables, F -values and ICC values.

4.3 Level of overall ethical work climate and its components

Table 7 shows the mean scores for overall EWC and its components. The mean score of the overall EWC is 5.19, indicating a moderate level of EWC in the NAD as perceived by the responding audit teams. In terms of the components of EWC, the mean scores range from 5.07 to 5.49. CM motivation has the highest mean value of 5.49, followed by CM sensitivity with a mean value of 5.43 and whistleblowing system with a mean value of 5.26. CM judgement has the lowest mean value of 5.07.

Secondly, to further explore the EWC and its components in the NAD, Table 9 presents the results on the average of each team's responses in percentage. Specifically, the mean of EWC and its components was calculated based on an audit teams' answers to all items (32 items) of EWC. If the mean of a team approximated "1", it was placed under scale "1"; if it approximated "2", it was categorized as scale "2"; and so on for the others. By doing this, the frequency of the team's responses could be calculated in percentage.

Table 9 shows the percentage of responses for mean values of EWC and its components from scale "1" up to scale "7". The scales of "1" to "7" were put into three groups: firstly, "1"

Table 8.
Means, standard
deviation, Pearson
bivariate correlations
and reliability

Variables	1	2	3	4	5	6	F(III)	ICC
1 Ethical work climate ^a	1						1.45***	0.078
2 CM sensitivity ^a	0.734**	1					1.24 *	0.036
3 CM judgement ^a	0.867**	0.483**	1				1.31*	0.052
4 CM motivation ^a	0.796**	0.325**	0.643**	1			1.36**	0.067
5 Whistleblowing system ^a	0.731**	0.652**	0.513**	0.338**	1		1.39**	0.068
6 OC ^b	0.796**	0.504**	0.720**	0.711**	0.499**	1	1.29*	0.047

Notes: CM: collective moral, OC: organizational commitment. p -value, $p < 0.05$, * $p < 0.01$, ** $p < 0.001$ ***(two-tailed). $N = 149$ teams, $N = 756$ individuals. ^a $N = 149$ teams. ^b $N = 765$ individuals

									Ethical work climate
Group Scales Percentage	1	Disagree 2 %	3	Neutral 4 %	5	Agree 6 %	7	Total	
EWC	0.00	0.00	0.00	5.40	38.20	54.40	2.00	100.00	733 Table 9. EWC and its components in Percentage of team means based on 7-point Likert scale
Collective	0.00	0.00	0.00	1.34	51.01	46.98	0.67	100.00	
Moral sensitivity	0.00			1.34	98.66				
collective	0.00	0.00	0.00	10.07	70.47	19.46	0.00	100.00	
Moral judgement	0.00			10.07	89.93				
Collective	0.00	0.00	0.00	4.70	38.93	51.00	5.37	100.00	
Moral motivation	0.00			4.70	95.30				
Whistleblowing	0.00	0.00	0.00	4.70	59.06	36.24	0.00	100.00	
system	0.00			4.70	95.30				

to “3” to generally reflect “disagree” to the NAD having a positive EWC; secondly, “4” as the mid-point, indicating neutrality, i.e. neither agreeing nor disagreeing that EWC in the NAD is positive or ethical; and thirdly, those with the team means between 5 and 7, reflecting “agree”.

By referring to the percentage mean values of teams for EWC for the “agree” group (category 5–7), the majority (94.60%) of audit teams concur that the work climate in the NAD are ethical. A minimal number of respondents (5.40%) are “neutral” about the overall EWC at the NAD. No audit team disagrees with an overall EWC in the NAD. The results imply an encouraging and positive sign in the NAD because the NAD is expected to maintain an EWC to perform its audit tasks.

In terms of analysing the percentage mean values of teams for the components of EWC of the “agree” group, public sector auditors have CM sensitivity (98.66%), CM motivation (95.30%), whistleblowing system (95.30%) and CM judgement (89.93%) at relatively high percentages. This is an indication that the audit teams agree most with the public sector auditors in the NAD being sensitive to a moral dilemma (CM sensitivity), followed by having whistleblowing system (CM motivation), being morally motivated (CM motivation) and finally, to having moral judgement (CM judgement).

4.4 Hypotheses testing: influence of ethical work climate and whistleblowing system on organizational commitment
Table 10 presents the HLM hypotheses testing results. *H1* predicted a positive association between EWC at the team level and the organizational commitment of public sector auditors at an individual level in the NAD. **Table 10** of Model 1 reveals that overall EWC positively and significantly predicts public sector auditors’ organizational commitment ($\gamma_{01}=0.41$, $SE = 0.02$, $p < 0.001$) in the NAD. This result is supported by a strong correlation at 79.60%, as presented in **Table 3**. This implies that the greater the EWC in the NAD, the more the public sector auditors are committed to their organization in the NAD. Thus, the finding supports *H1*. The result is in line with empirical results of [Wimbush et al. \(1997\)](#), [Vardi \(2001\)](#), [Peterson \(2002\)](#) and [Weber et al. \(2003\)](#), as they found the stronger the EWC of an organization, the lower the negative behaviour of employees or in other words, more positive behaviour. However, these prior studies were performed at a single level of analysis. This finding also supports [Arnaud’s \(2010\)](#) EWC model that predicts that positive EWC

leads to positive outcomes; in this case, the positive outcomes refer to organizational commitment.

The study further explored the influence of each component of EWC on public sector auditors' organizational commitment using multivariate cross-level effect tests. The four components of EWC – CM sensitivity (*H2*), CM judgment (*H3*), collectively moral motivation (*H4*) and whistleblowing system (*H5*) were hypothesized as having a positive influence on the organizational commitment of public sector auditors in the NAD. To examine *H2* to *H5*, lower level organizational commitment was regressed to the upper level of the four components of EWC, respectively.

As shown in [Table 10](#), three of the EWC components have a positive significant effect on organizational commitment: CM sensitivity ($\gamma_{01}=0.11$, standard error = 0.04, $p < 0.001$) and CM judgement ($\gamma_{01}=0.16$, SE = 0.03, $p < 0.001$), whereas CM motivation ($\gamma_{01}= - 0.20$, SE = 0.03, $p < 0.001$) has a negatively significant result. Whistleblowing system ($\gamma_{01}=0.14$, SE = 0.04, $p < 0.01$) also shows a positive significant cross-level effect on public sector auditors' organizational commitment in the NAD context. Specifically, the findings can be interpreted as follows: when audit team members possess high moral thinking and evaluation when facing ethical challenges (CM sensitivity) and have an ethical judgement (CM judgement), they are led to be committed to their organization. Thus, *H2*, *H3* and *H5* are supported, whereas *H4* is not supported. The results are in line with empirical results of [Wimbush et al. \(1997\)](#), [Vardi \(2001\)](#), [Peterson \(2002\)](#) and [Weber et al. \(2003\)](#), as they found the stronger the EWC of an organization, the lower the negative behaviour of employees, i.e. more positive behaviour. However, these studies were performed at a single level of analysis. This finding also supports [Arnaud's \(2010\)](#) EWC model that predicts that positive EWC leads to positive outcomes; in this case, the positive outcome refers to organizational commitment.

Interestingly, CM motivation has a significantly negative effect on organizational commitment. This seems to indicate that organizational commitment depends on CM sensitivity and CM judgement because if public sector auditors are sensitive to ethical dilemmas and their ethical judgement is sound, it would affect their organizational commitment positively. Having ethical concern to do the right thing does not necessarily result in the motivation to do the right thing. In fact, in the context of this study's data, having ethical concerns about doing the right thing could still result in negative organizational commitment. Another possible explanation for the significant negative influence of CM motivation is that audit teams working in an environment characterized by a negative level of motivation may still be likely to show organizational commitment due to their high level of CM sensitivity and CM judgement. In fact, it could be possible that if the audit team has high collective sensitivity and CM judgement, CM motivation will be considered a less significant influence, as organizational commitment is still achieved. This is an

Table 10.
Summary result of
HLM hypotheses
testing

Model	Cross-level effects	Organizational commitment
1	H1 EWC	0.41 (0.02)***
2	H2 collective moral sensitivity	0.11 (0.04)**
3	H3 collective moral judgment	0.16(0.03) ***
4	H4 collective moral motivation	−0.20(0.03)***
5	H5 whistleblowing system	0.14(0.04)**

Notes: DV: dependent variable, CM: collective moral. The first value is the parameter estimate and the value within parenthesis is the SE. p -value, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

interesting finding because it shows that not all components of EWC work in the same direction to produce positive outcomes as expected from [Arnaud's \(2010\)](#) and [Rest's \(1994\)](#) models. This is in spite of overall EWC having positive outcomes on organizational commitment.

Logically, the researcher of the present study thinks that an individual would experience the four components of EWC mentally, which are subjective and unobservable. It is possible for an individual not to proceed with all components to obtain organizational commitment. A person can be motivated to act unethically but may eventually take ethical action. As applied to this study, public sector auditors in the NAD would be committed if they experience high levels of moral sensitivity and moral judgment, even with significantly low moral motivation. This result highlights that as far as the public sector auditors can reach the moral judgment stage, they could be committed, even if other concerns (such as power and self-achievement) prevail. The HLM finding supports that it is not necessary to display all four components of EWC to become an ethical person.

5. Implications of the study

The present study explores the level of EWC of public sector auditors and examines the relationship between overall ECW and its components and organizational commitment of public sector auditors at the NAD. A cross-sectional questionnaire survey was distributed to the population of the public sector auditors in Malaysia, and data was analysed at different levels of measurement. The findings of the study reveal that the level of EWC and its components at the NAD is at a reasonable level, i.e. with an average mean score of more than 5 out of 7 on the scale. The study also finds that the overall EWC at the team level has a positive significant influence on organizational commitment assessed at the individual level of auditors. In terms of the individual components of the EWC, CM sensitivity and CM judgement have significantly positive relationships with organizational commitment, whereas CM motivation has a negatively significant relationship with organizational commitment. On the other hand, there is a significant relationship between whistleblowing system and organizational commitment.

There are several implications from the findings of the present study. In terms of the practical implications, first, the empirical evidence of this present study on the moderate level of EWC and its components implies that there is room for improvement in the ethical climate in the NAD. The finding is useful to the authorities, particularly the AG's office, to design necessary measures to enhance the EWC, including tightening the auditors' code of ethics and organizing ethics-related seminars, training and workshops for the auditors. Moreover, the NAD may also want to consider imposing higher requirements in terms of continuous professional development hours for auditors. In inculcating the EWC in the NAD, creative measures, such as campaigns, by displaying attractive posters on the NAD premises and disseminating interactive animation posters via email on the dos and don'ts at work as well as the implications of engaging in unethical work conduct, can be carried out.

Secondly, the result on the positive influence of CM sensitivity, CM judgment and whistleblowing system on organizational commitment of public sector auditors is also crucial for the NAD. It must take measures to further inculcate moral sensitivity and moral judgment among the public sector auditors to ensure a continuously positive relationship with the organizational commitment of auditors, which ultimately will assist in restoring the credibility of the accounting and auditing profession that has been, to some extent, tarnished due to unsatisfactory ethical behaviour among auditors. Thirdly, in relation to the negative influence of CM motivation on organizational commitment, it offers an avenue for ethics experts and researchers to further explore the relationship as it could be possible

that there are other elements/variables that significantly mediate the relationship, and having an appropriate mediating variable may change the direction of the relationship.

In relation to theoretical implications, this study provides empirical support for combining Hunt and Vitell's (1991) theory of marketing ethics and Arnaud's (2010) PPM in examining the level of EWC and outcomes of EWC. Overall, the findings of the study support the hypotheses developed in accordance with Hunt and Vitell's and Arnaud's models. Arnaud's (2010) model is, in turn, supported by Rest's (1994) model. As explained earlier, Arnaud's model is based on a collective perspective, whereas Rest's model refers to an individual perspective. The combination of models in a study may allow future researchers to develop different research coverage. In the context of the NAD in Malaysia, all components in a psychological process of making ethical decisions – CM sensitivity, CM judgement, CM motivation, whistleblowing system – together create EWC. Thus, future researchers need to include all four components in their studies. Moreover, consistent with Rest's and Arnaud's models, all EWC components influence organizational commitment; however, inconsistent with these two models, not all these components generate positive outcomes.

6. Limitations, suggestions for future studies and conclusion

This study has limitations. Firstly, this research is a cross-sectional study. The problem arises when data are collected at a single point in time, which may cause common method variance bias to arise. However, the limitation can be reduced by using mixed research methods or conducting a longitudinal study; these methods were not used in this study. This study performed Harman's single factor test and used a multi-level approach to alleviate the possibility of common method variance. Therefore, it is recommended that future studies adopt a mixed research method to explore EWC and confirm the findings of this present study. Future researchers may use mixed methods to capture better insights into the EWC in all types of organizations, for example, questionnaire and interview, questionnaire and focus group or questionnaire and observation.

Secondly, the study used a survey method to gather data from numerous respondents in a largely objective manner. However, the quantitative method, unlike the qualitative method, may have failed to provide richer data of EWC in the NAD. Such rich, in-depth data on ethical practices of the public sector auditors may be obtained through observation and interview methods. Therefore, future studies may investigate EWC through a mixed-method approach, such as a combination of observation or interview and questionnaire.

Finally, another limitation of this research is generalizing the findings to other organizational contexts, for instance, private sector audit firms, as this study was conducted on public sector auditors in the NAD. However, the findings can be generalized across the public sector auditors in the NAD. This is because this study did not just collect data from lower level (position) public sector auditors but also from auditors at multi-levels (including the top management). Therefore, it is strongly recommended that future studies explore the EWC and its outcomes in different contexts, such as in accounting, tax, marketing, agriculture, construction, food industry and culture contexts.

Despite the mentioned limitations, this study offers relevant information to stimulate more interest in this area of research and paves the way for future studies. More importantly, the implications from the findings of the present studies are useful to the various stakeholders, including the NAD, AG, public sector auditors, as well as the experts and researchers in the area of ethics and auditing. The results may also support the NAD in the development of an EWC. Such effort is desirable because an EWC is needed in the audit profession to reduce financial scandals and to ensure the accountability and reliability of the

NAD. The present study may also help to improve the credibility of the accounting and auditing profession, which has been damaged due to the misconduct of some auditors.

Ethical work
climate

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Corresponding author

Suhaiza Ismail can be contacted at: suhaiza@iium.edu.my